

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S8 (R,S) Exam (FT / PT) April 2025 (2019 Scheme)

Course Code: CST448

Course Name: INTERNET OF THINGS

Max. Marks: 100**Duration: 3 Hours**

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|--|-----|
| 1 | List the most significant challenges in IoT and briefly explain about any two. | (3) |
| 2 | Compare OT and IT technology. | (3) |
| 3 | Discuss the limitations of Smart objects in Wireless Sensor Network. | (3) |
| 4 | Write a short note on MEMS with an example. | (3) |
| 5 | Discuss the role of mesh addressing in 6LoWPAN. | (3) |
| 6 | List advantages of Internet Protocol (IP). | (3) |
| 7 | Explain | |
| | a. Supervised Learning | (3) |
| | b. Unsupervised Learning | |
| 8 | Differentiate between structured and unstructured data. | (3) |
| 9 | Discuss the significance of python programming IoT development. | (3) |
| 10 | Explain the role of RESTful APIs in IoT system integration. | (3) |

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- | | | |
|----|---|-----|
| 11 | a) Detail about the IoT Reference Model Published by the IoT World Forum. | (9) |
| | b) Describe the evolutionary phases of Internet. | (5) |

OR

- | | | |
|----|--|-----|
| 12 | a) With a neat diagram, explain the one M2M IoT Standardized Architecture. | (8) |
| | b) Describe the defining characteristics of fog computing. | (6) |

Module II

- 13 a) Explain the communication criteria which are to be considered when selecting and dealing with connecting smart objects. (8)
- b) Explain any three advantages and disadvantages of SANET. (6)

OR

- 14 a) Explain the Characteristics of smart objects. Also list and explain the trends in smart objects. (10)
- b) List and explain any three protocol stacks utilizing IEEE 802.15.4. (4)

Module III

- 15 a) Describe the schedule management mechanism and packet forwarding models in 6TiSCH. (9)
- b) How COAP differ from MQTT. (5)

OR

- 16 a) Discuss the need for optimization of IP in IoT. (6)
- b) Write a short note on Message Queuing Telemetry Transport (MQTT). (8)

Module IV

- 17 a) Illustrate the common challenges in OT Security. (6)
- b) Explain with a block diagram the types of Data analysis. (8)

OR

- 18 a) Demonstrate Formal Risk Analysis structure Octave. (6)
- b) Write a short note on Flexible Net-Flow Architecture (FNF). (8)

Module V

- 19 a) Explain the Django Architecture. (7)
- b) Demonstrate an example of Raspberry Pi application for industrial IoT (7)

OR

- 20 a) Describe how AWS supports IoT development. (7)
- b) Explain the working of WAMP protocol. (7)
